Goutham
AI Engineer | Gen AI Engineer | ML Engineer | Prompt Engineer |
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Professional Summary

- 9+ years of experience in AI/ML, Generative AI, NLP, MLOps, and Big Data Engineering across healthcare, banking, entertainment, telecom, and finance domains.
- Expertise in designing, developing, and deploying end-to-end AI/ML solutions across Azure, GCP, AWS, and Big Data ecosystems for scalable and cost-efficient applications.
- Proven experience in Large Language Models (LLMs), including GPT (OpenAI), BERT, Gemini (Google), Llama (7B, 13B, 34B), Groq, Phi-Data, Claude (Anthropic), Mistral, Falcon, and vector search applications for knowledge retrieval and conversational AI.
- Designed and implemented Retrieval-Augmented Generation (RAG)-based AI Agents using LangChain, LangFlow, and Vector Databases.
- Hands-on experience in MLOps with MLflow, Kubeflow, Azure DevOps, Jenkins, and CI/CD automation pipelines for production-ready ML models.
- Expertise in NLP-based AI applications, including text classification, entity recognition, summarization, and intelligent chatbots for enterprise applications.
- Deep knowledge of cloud-based AI services, including Azure ML, AWS SageMaker, GCP Vertex AI, and Hugging Face for scalable model training and inference.
- Experience with real-time and batch data processing frameworks, including Apache Spark, Flink, Kafka, and Google Dataflow, to optimize AI-driven workflows.
- Strong background in data engineering and ETL pipeline orchestration using Apache Airflow, AWS Glue, and Azure Data Factory for seamless data transformations.
- Expertise in AI-driven microservices architecture using Kubernetes, Docker, REST APIs, GraphQL, and event-driven architectures.
- Experience in GPU/TPU acceleration for deep learning models and optimizing inference latency for real-time AI applications.
- Knowledge of SDLC, Agile/Scrum methodologies, and DevOps best practices, ensuring rapid development, iteration, and deployment of AI solutions.
- Strong focus on AI governance, data privacy, and regulatory compliance (HIPAA, GDPR, SOC 2, and FDA) to build trustworthy AI applications.
- Proven leadership in mentoring junior AI engineers, conducting AI research, and implementing best practices in AI/ML model development.
- Collaborative experience with cross-functional teams (product managers, data scientists, engineers) to deliver AI-driven solutions that drive business impact.

Technical Skills

Category	Skills
Machine Learning & GenAI	TensorFlow, PyTorch, Scikit-learn, MLflow, Hugging Face, JAX, OpenAI API, LangChain, Llama Models (7B, 13B, 34B), RAG-based AI, Fine-tuning LLMs
Prompt Engineering	Chain-of-Thought (CoT), Few-Shot Learning, Zero-Shot Learning, LLM Optimization, Prompt Tuning
Programming Languages	Python (2.x/3.x), Java, Scala, R, PySpark, PL/SQL, Shell Scripting, PowerShell
Data Engineering	Apache Spark (Core, SQL, MLlib, Streaming), Kafka, Hive, HDFS, Airflow, AWS Glue, Azure Data Factory
Cloud Platforms	AWS (SageMaker, Redshift, Lambda, Glue), Azure (Databricks, Azure ML, Synapse, ADLS), GCP (Vertex AI, BigQuery, Pub/Sub)

Statistical Methods	Regression Models, Decision Trees, Ensemble Models, ANOVA, PCA, Time-Series (ARIMA, GARCH), Hypothesis Testing
MLOps & Automation	MLflow, Kubeflow, TensorBoard, Jenkins, Azure DevOps, Docker, Kubernetes, Terraform, and Ansible
Streaming and Orchestration, Scheduling and Container Management	Apache Kafka, Apache Flink, Google Dataflow, Apache Airflow, AWS Step Functions, Kubernetes workflows, GitOps, and MLflow
Data Visualization	Power BI, Tableau, Google Data Studio, Matplotlib, Seaborn, Plotly, R Shiny
Big Data & Storage	HDFS, Snowflake, Amazon Redshift, BigQuery, Azure Synapse Analytics
Data Governance & Security	Azure AD, AWS IAM, Apache Atlas, Ranger, Sentry, Key Vault
Version Control	Git, Bitbucket, GitHub, SVN
Monitoring Tools	Splunk, AWS CloudWatch, Azure Monitoring, Google Stackdriver

Professional Experience

Senior AI/ML Engineer, **Client: CVS – Florham Park, NJ**
Jan 2024 – Present
Tech Stack: Azure ML, Synapse, Vector DBs, LangChain, OpenAI API, Kubernetes, Docker, Azure Data Factory

- Worked on LLM-powered AI agent deployments for healthcare automation, focusing on enhancing medical claim processing and fraud detection.
- Designed RAG architectures using LangChain, LangFlow, and vector databases for knowledge retrieval.
- Developed predictive analytics models for patient risk assessments and fraud detection.
- Automated CI/CD pipelines for ML models using MLflow, Kubeflow, and Azure DevOps.
- Optimized cloud-native AI microservices using Azure Kubernetes Service (AKS) and Docker.
- Built ETL workflows for structured/unstructured medical data (FHIR, HL7, SQL).
- Implemented model inference acceleration using GPUs and TPUs.
- Ensured HIPAA, GDPR, and FDA compliance for AI-driven applications.
- Designed secure multi-tenant AI solutions for enterprise healthcare platforms.
- Integrated AI-powered medical automation workflows to improve operational efficiency.

AI Engineer / ML Engineer **Client : Mastercard, St Louis,Mo**
Oct 2022 – Dec 2023
Tech Stack: GCP, Vertex AI, BigQuery, Kubeflow, TensorFlow, PyTorch, BERT, GPT, LSTMs

- Developed real-time fraud detection models leveraging transformer architectures.
- Deployed credit risk prediction models using GCP Vertex AI & AutoML.
- Created AI-powered chatbots using LangChain & OpenAI API.
- Built scalable AI inference pipelines using Kubeflow & Google Cloud Functions.
- Optimized time-series forecasting techniques for anomaly detection.
- Integrated AI-based document summarization into enterprise workflows.
- Designed MLOps workflows for model versioning and continuous training.
- Benchmarked AI model performance & hyperparameter tuning.
- Developed BigQuery-based AI monitoring dashboards.
- Implemented AI-powered user authentication & risk assessment models.

ML Engineer/ Python Developer **Client : Sony Music Entertainment, Rutherford, NJ**
Jan 2022 - Sep 2022
Tech Stack: AWS, SageMaker, Lambda, PyTorch, TensorFlow, NLP, FastText, Spacy

- AI-powered music recommendation systems have been developed using collaborative filtering and deep learning techniques.

- Text classification models have been deployed for genre tagging, sentiment analysis, and lyric categorization.
- Real-time audio transcription pipelines have been implemented using deep learning-based speech-to-text models.
- Generative AI-based techniques have been explored for music composition and augmentation.
- AI-driven data analytics dashboards have been designed to provide insights into music streaming trends.
- LLM-based AI solutions have been integrated to enhance fan engagement and chatbot-driven interactions.
- MLOps workflows on AWS SageMaker have been automated for continuous model retraining and deployment.
- Flask-based AI APIs have been developed to facilitate the integration of ML models with streaming services.
- Natural language processing techniques have been applied to analyze consumer music preferences and behavior.
- AI-driven personalized music recommendation systems have been optimized to improve user engagement and retention.
- A knowledge retrieval system has been developed to provide actionable insights into the music industry.
- Artist contract analysis has been automated using AI-powered NLP models to enhance efficiency and accuracy.

Data Engineer, **Client: Value Labs, Hyderabad, Telangana**
Jan 2018 - July 2021
Tech Stack: Azure (Data Lake, Synapse, HDInsight, Blob Storage, Data Factory, Purview), Apache Spark, Scala, HDFS, Apache Kafka, Hive, Airflow, SQL Server, Delta Lake, Tableau.

- Big data pipelines were designed and optimized for processing large-scale datasets using Azure Data Lake, HDFS, and Apache Spark.
- Real-time data ingestion frameworks were implemented utilizing Apache Kafka, Azure Event Hubs, and Azure Stream Analytics for streaming data processing.
- ETL workflows were developed and orchestrated using Apache Spark, Hive, and Azure Data Factory to enable scalable data transformations.
- Data integration pipelines were established to process structured, semi-structured, and unstructured data sources using Azure Synapse Analytics and Apache Hive.
- Storage and data lake architectures were optimized using HDFS, Delta Lake, and Azure Blob Storage to enhance data organization and retrieval efficiency.
- Data governance frameworks were enforced with Azure Purview, Apache Ranger, and role-based access controls to ensure compliance and security policies.
- Workflow automation was configured using Apache Airflow and Azure Logic Apps for seamless job scheduling and orchestration.
- Query performance optimization techniques were applied using Azure Synapse, SQL Server, and Hive to improve data retrieval and processing speeds.
- Collaborative efforts were undertaken with data analysts and business teams to derive AI-driven insights using Tableau and Azure Synapse.
- On-premises Hadoop workloads were migrated to Azure HDInsight and Synapse Analytics, improving cost-efficiency, scalability, and processing performance.
- Streaming analytics solutions were integrated with ML models on Azure Machine Learning and Kafka Streams for real-time anomaly detection and predictive modeling.
- Cost-efficient, cloud-native data architectures were designed using Azure-based Hadoop ecosystems, Delta Lake, and Spark Structured Streaming to support AI-driven data processing at scale.

Data Analyst, **Client: EKA Analytics, Bangalore, India**
April 2015 - Dec 2017
Tech Stack: Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-learn), SQL, Google Cloud Platform (BigQuery, Cloud Storage), Excel

- Analytical tools were developed using Python (Pandas, NumPy) and SQL to streamline data processing and analysis.
- Loan data was collected, cleaned, and analyzed using SQL and Python, enabling cross-validation and credit risk assessment.
- Statistical data models were designed using Python (SciPy, Scikit-learn), SQL, and Excel, contributing to a 3% reduction in bad debts.
- Machine learning models were validated for credit risk assessment (Probability of Default) using Python and BigQuery ML.
- Customer, account, and transaction data APIs were tested to ensure accuracy and consistency in financial analytics workflows.
- Credit check procedures were refined, leading to the elimination of 15% of unqualified applicants before loan underwriting.
- Automated query pipelines were developed using Python and SQL, reducing query resolution time by 45%.
- Data visualization dashboards were created using Matplotlib, Seaborn, and Google Data Studio to support business intelligence and reporting.
- Actionable insights and recommendations were derived through data interpretation and statistical analysis.
- Data cleaning, transformation, and organization were performed using SQL and Python (Pandas, NumPy) to enhance data accuracy and efficiency.
- Reports and trend analysis dashboards were developed using Google BigQuery, SQL, and Python visualization libraries to support data-driven decision-making.
- Collaboration with stakeholders was undertaken to gather business requirements and optimize data analytics workflows for improved reporting and forecasting.

Education

- Master’s in Information Systems -- USA
- Bachelor’s in Electrical Engineering – India